

Efficacy of short-term monotherapy with maraviroc, a new CCR5 antagonist, in patients infected with HIV-1

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We assessed the efficacy and safety of 10-d monotherapy with the orally administered CCR5 antagonist maraviroc in 63 HIV-1-positive individuals prescreened for the absence of CXCR4-using virus. Maximum reduction in viral load occurred at a median of 10–15 d, with a mean reduction of $\geq 1.6 \log_{10}$ copies/ml at all twice daily doses ≥ 100 mg. These results provide proof of concept that CCR5 antagonism is a viable antiretroviral therapeutic approach.

The early success of highly active antiretroviral therapy has been tempered by the emergence of cross-resistance, side effects, long-term toxicity and complex dosing regimens¹, and there remains a need for well-tolerated, conveniently administered agents with new mechanisms of action. The HIV coreceptor CCR5 is of particular interest as a drug target because a natural mutation (CCR5- $\Delta 32$) leads to reduced or absent expression of CCR5 in heterozygous or homozygous genotypes, respectively, with no apparent deleterious consequences. Individuals homozygous with respect to CCR5- $\Delta 32$ have a high degree of resistance to HIV infection, whereas heterozygotes have a reduced rate of disease progression^{2–4}. CCR5 antagonists represent a promising new class of entry inhibitors under development for the treatment of HIV-1 infection^{5–8}. The CCR5 antagonist maraviroc has potent anti-HIV activity *in vitro*⁹ and is safe and well-tolerated at multiple doses up to 300 mg twice daily (BID) in healthy volunteers (M. McHale, S. Abel, D. Russell, J. Gallagher and E. van der Ryst, unpublished results).

We carried out two randomized, placebo-controlled, phase 2a studies (A4001007 and A4001015) to investigate the antiviral efficacy of short-term maraviroc monotherapy in asymptomatic, CCR5-tropic HIV-1-infected patients. Because the study designs (**Supplementary**

Methods online) and patient populations were similar, we combined the results to gain a clearer understanding of the dose-response relationship. Of 163 patients screened, 82 entered the studies. The most common reasons for exclusion were evidence of hepatic impairment (12 of 81), nonphenotypable virus (10 of 81) and a CD4⁺ cell count of <250 cells/mm³ (10 of 81). Seven patients were excluded owing to detection of dual- or mixed-tropic virus. The overall prevalence of CCR5-tropic virus in screened patients for whom a final tropism result was available was 94% (109 of 116). Patients were randomized to receive 10 d of treatment with maraviroc (25 mg, 100 mg or 300 mg once daily (QD) or 50 mg, 100 mg, 150 mg (fed and fasted) or 300 mg BID) or placebo (**Table 1**). Baseline demographics and disease characteristics were similar across all treatment groups, with mean baseline viral load and CD4⁺ counts of 4.62 (range 3.56–5.64) $\log_{10}$ HIV RNA copies/ml and 544 (range 205–1137) cells/mm³, respectively.

All maraviroc treatment groups had a greater mean decrease in viral load, from baseline to day 11, than did placebo groups (**Table 1** and **Fig. 1a**). Once treatment was discontinued, viral load rebound was not immediate; maximum viral load reduction occurred after day 11 in several patients (median time to nadir = 10–15 d; **Table 1**). At doses of 100 mg BID and above, all patients, with the exception of one patient with dual- or mixed-tropic virus at baseline (who was inadvertently enrolled owing to a sample-switching error and who experienced no overall reduction in viral load), experienced a reduction in viral load of $>1.0 \log_{10}$ copies/ml at nadir (**Table 1**). Mean CD4⁺ count changes between day 1 and day 11 in the maraviroc treatment groups were variable, and there was no correlation between changes in CD4⁺ count and viral load response or dose (**Supplementary Note** online).

The inconvenience of frequent dosing or food restrictions can represent a barrier to treatment adherence¹⁰. We compared BID and QD dosing and the effect of food on response to treatment. There was no difference in reduction in viral load at day 11 between groups treated with 150 mg BID versus 300 mg QD or between groups treated with 150 mg BID that were fed versus fasted. At 150 mg BID, food reduced the $C_{\max}$ and AUC_{τ} by $\sim 60\%$ and $\sim 50\%$, respectively, but did not seem to alter the $C_{\min}$ (**Supplementary Table 1** online). Viral load reductions on maraviroc were similar at all doses of 100 mg BID and above, seemed to be close to the maximum on the systemic exposure (AUC) response curve, and were comparable to those observed with potent protease inhibitors and efavirenz^{11–13} (**Fig. 1b**). Taken together, these results imply that maraviroc is likely to be suitable for QD dosing without food restrictions.

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Table 1 Changes in viral load from baseline

Dose	Number of patients		Median time to nadir, d	Mean change in HIV-1 RNA, log ₁₀ copies/ml (range)		Number of patients with >1.0 log ₁₀ decrease in HIV-1 RNA	
	Randomized	Treated for 10 d		Day 11	Nadir	Day 11	Nadir
25 mg QD	9	8 ^a	10	-0.43 ^e (-1.08, 0.02)	-0.59 (-1.10, 0.02)	1	1
50 mg BID	8	8	13	-0.66 ^d (-1.37, 0.40)	-0.86 (-1.37, -0.14)	4	5
100 mg QD	9	8 ^b	10	-1.13 (-1.70, -0.43)	-1.25 (-1.70, -0.61)	5	6
100 mg BID	8	7 ^c	12	-1.42 ^d (-1.84, -1.04)	-1.68 (-2.10, -1.37)	7	7
150 mg BID	8	8	12	-1.45 ^{g,h} (-1.71, -0.90)	-1.77 (-2.16, -1.43)	7	8
150 mg BID (fed)	8	8	15	-1.34 ^h (-1.79, -0.51)	-1.74 (-2.09, -1.13)	7	8
300 mg QD	8	8	12	-1.35 ^g (-1.62, -0.95)	-1.60 (-2.08, -1.14)	7	8
300 mg BID	8	8	11	-1.60 ^f (-2.42, -0.78)	-1.84 (-2.42, -1.49)	7	8
Placebo (1007)	12	12	8 ⁱ	0.02 (-0.45, 0.56)	-0.32 (-0.63, 0.11) ⁱ	0	0 ⁱ
Placebo (1015)	4 ^d	4		0.09 (-0.20, 0.27)		0	

^aOne patient withdrew consent. ^bOne patient discontinued treatment because of a moderate headache (not considered to be treatment related) before dosing on day 3. ^cOne patient in the group treated with 100 mg BID had dual- or mixed-tropic virus at baseline and was excluded from the efficacy analysis. Inclusion of this patient in the analysis resulted in a mean change in HIV-1 RNA from baseline to day 11 of -1.20 log₁₀ copies/ml (range -1.84 to 0.33). ^dTwo fed and two fasted. ^e $P = 0.056$ compared with placebo, Williams step-down test. ^f $P < 0.01$ compared with placebo, Williams step-down test. ^gMean difference between 150 mg BID and 300 mg QD was -0.10 log₁₀ copies/ml; 90% confidence interval was -0.39, 0.19. ^hMean difference between 150 mg BID fed and 150 mg BID fasted was -0.10 log₁₀ copies/ml; 90% confidence interval was -0.39, 0.19. ⁱThese values apply to both placebo groups.

Analysis of maraviroc pharmacokinetic data indicated that absorption was rapid but variable, with $t_{\max}$ generally occurring between 1 and 4 h after dosing (Supplementary Table 1). There was no evidence of maraviroc accumulation at any dose when comparing AUC₀₋₁₂ after single dosing (day 1) with that at steady state after multiple dosing (day 10; Supplementary Table 1). Because maraviroc does not act on a viral target but rather is an antagonist of a human cellular receptor, we also evaluated occupancy of CCR5, determined using an experimental MIP-1 β internalization assay (Supplementary Methods). All maraviroc treatment groups had mean predose CCR5 occupancy of >80% on day 5, and for all doses except 25 mg QD this remained >80% on day 10 predose and >60% on day 15. The delayed rebound in viral load after treatment discontinuation, which has also been described for other CCR5 antagonists¹⁴, may be due to prolonged occupancy of CCR5 by maraviroc after treatment had been discontinued. However, no association between degree of CCR5 occupancy and viral load reduction was apparent. A possible explanation for this observation is that very high levels of receptor occupancy are needed for optimal antiviral activity, and that the inherent variability of the experimental assay may

result in an inability to detect small differences in receptor saturation. Furthermore, the assay measures only peripheral blood receptor saturation and gives no indication of the degree of tissue receptor occupancy. The relationship between *ex vivo* receptor occupancy, as measured by this assay, and functional *in vivo* receptor occupancy is therefore unclear. Finally, the degree to which prolonged CCR5 occupancy delays the rebound in viral load may also be related to the plasma half-life of the drug, as unbound CCR5 molecules are continuously being regenerated *in vivo*.

Concern exists that selective pressure from a CCR5 antagonist could lead to early predominance of CXCR4-using strains in patients infected with mixed- or dual-tropic HIV. We therefore assessed viral tropism at screening and on days 1, 11 and 40 for all patients. Changes in virus tropism were seen in two patients receiving maraviroc 100 mg QD, both of whom had viral load responses similar to those of other patients in the treatment group. In one patient there was a transient emergence of dual- or mixed-tropic virus on day 11, but on day 40 only CCR5-tropic virus could be detected. In the second patient, dual- or mixed-tropic virus was detected on day 11 and remained detectable at follow-up

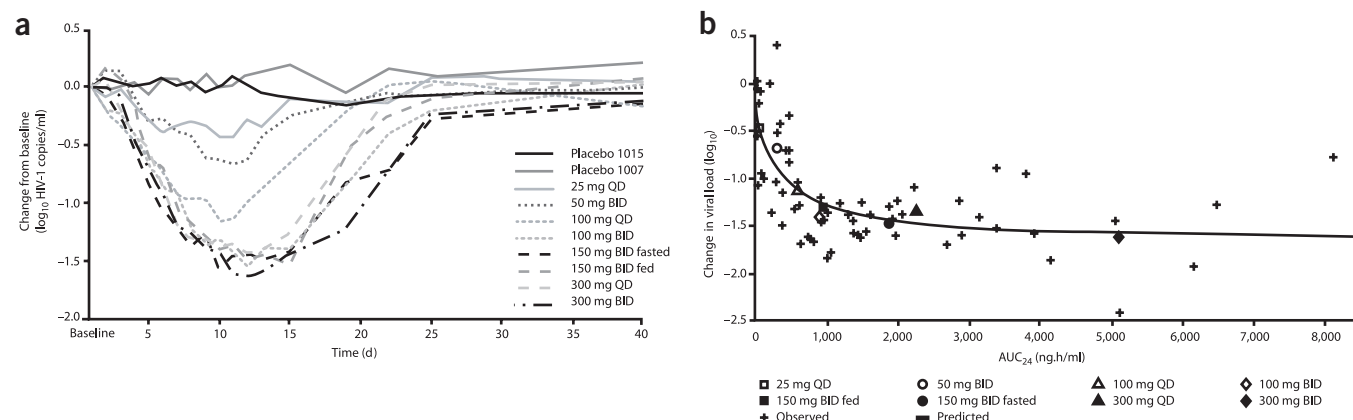


Figure 1 Change in viral load over time and its correlation with maraviroc steady state AUC₀₋₂₄. **(a)** Mean change in log₁₀ plasma viral load over time by dose group. **(b)** Maraviroc steady state AUC₀₋₂₄ versus change in viral load from baseline on day 11. Crosses represent the observed values for each individual patient; other symbols represent the mean values for each treatment group. An $E_{\max}$ model was fitted, using WinNonlin Professional version 4.1, to describe the relationship between steady state AUC₀₋₂₄ and log₁₀ reduction in viral load on day 11 (solid line). The model predicted a maximum drop in viral load of 1.44 ± 0.20 (standard error) log₁₀ copies/ml; the AUC₀₋₂₄ required to achieve 50% of this reduction was 353 ± 196 (standard error) ng h/ml.

to day 433. A more detailed description of the analysis of virus tropism in these patients will be published elsewhere (M. Westby *et al.*, unpublished data). These results suggest that CCR5 tropism is strongly selected for in patients, even following the presence of a strong counter-selective pressure. Only long-term studies of CCR5 antagonists, in combination with other antiretroviral drugs, in patients with dual- or mixed-tropic virus will further address this issue.

Maraviroc was well-tolerated at all doses in these studies. The only treatment-emergent adverse events that occurred more than once in any dose group (including placebo) were headache, asthenia, dizziness, gingivitis and nausea (**Supplementary Note**).

In summary, these results show that the CCR5 receptor is a valid, nonviral target for the treatment of HIV-1-infected individuals and indicate that further clinical evaluation of maraviroc is merited. Similar reductions in viral load have been observed after short-term monotherapy with other investigational drugs in this class¹⁴. Given the lack of cross-resistance to other classes and the high prevalence of CCR5 tropic virus in HIV-infected patients, this compound may provide a valuable addition to the armamentarium of antiretroviral drugs.

Note: Supplementary information is available on the Nature Medicine website.

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COMPETING INTERESTS STATEMENT

The authors declare competing financial interests (see the *Nature Medicine* website for details).

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1. Struble, K. *et al.* *AIDS* **19**, 747–756 (2005).
2. Dean, M. *et al.* *Science* **273**, 1856–1862 (1996).
3. Eugen-Olsen, J. *et al.* *AIDS* **11**, 305–310 (1997).
4. Meyer, L. *et al.* *AIDS* **11**, F73–F78 (1997).
5. Kuhmann, S.E. *et al.* *J. Virol.* **78**, 2790–2807 (2004).
6. Maeda, K. *et al.* *J. Virol.* **78**, 8654–8662 (2004).
7. Strizki, J.M. *et al.* *Proc. Natl. Acad. Sci. USA* **98**, 12718–12723 (2001).
8. Takashima, K. *et al.* *Antimicrob. Agents Chemother.* **45**, 3538–3543 (2001).
9. Dorr, P. *et al.* *Antimicrob. Agents Chemother.* (in the press).
10. Maggiolo, F. *et al.* *HIV Clin. Trials* **3**, 371–378 (2002).
11. Adkins, J.C. & Noble, S. *Drugs* **56**, 1055–1064 (1998).
12. Gulick, R.M. *et al.* *N. Engl. J. Med.* **337**, 734–739 (1997).
13. McCallister, S. *et al.* *J. Acquir. Immune Defic. Syndr.* **35**, 376–382 (2004).
14. Lalezari, J. *et al.* *AIDS* **19**, 1443–1448 (2005).